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Surgical stapler with universal tip reload

Abstract

A surgical stapler includes a tool assembly that has a cartridge assembly, an anvil assembly, and a dissector tip. The dissector tip is moveable between a first configuration aligned with a longitudinal axis of the tool assembly and a second configuration defining an acute angle with the longitudinal axis of the tool assembly.

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6981628	12/2005	Wales	N/A	N/A
6986451	12/2005	Mastri et al.	N/A	N/A
6988649	12/2005	Shelton, IV et al.	N/A	N/A
6991627	12/2005	Madhani et al.	N/A	N/A
6994714	12/2005	Vargas et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7000819	12/2005	Swayze et al.	N/A	N/A
7032799	12/2005	Viola et al.	N/A	N/A
7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044353	12/2005	Mastri et al.	N/A	N/A
7055730	12/2005	Ehrenfels et al.	N/A	N/A
7055731	12/2005	Shelton, IV et al.	N/A	N/A
7059508	12/2005	Shelton, IV et al.	N/A	N/A
7070083	12/2005	Jankowski	N/A	N/A
7083075	12/2005	Swayze et al.	N/A	N/A
7097089	12/2005	Marczyk	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7114642	12/2005	Whitman	N/A	N/A
7121446	12/2005	Arad et al.	N/A	N/A
7128253	12/2005	Mastri et al.	N/A	N/A
7128254	12/2005	Shelton, IV et al.	N/A	N/A
7140527	12/2005	Ehrenfels et al.	N/A	N/A
7140528	12/2005	Shelton, IV	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7143924	12/2005	Scirica et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7143926	12/2005	Shelton, IV et al.	N/A	N/A
7147138	12/2005	Shelton, IV	N/A	N/A
7159750	12/2006	Racenet et al.	N/A	N/A
7168604	12/2006	Milliman et al.	N/A	N/A
7172104	12/2006	Scirica et al.	N/A	N/A
7188758	12/2006	Viola et al.	N/A	N/A
7207471	12/2006	Heinrich et al.	N/A	N/A
7213736	12/2006	Nales et al.	N/A	N/A
7225963	12/2006	Scirica	N/A	N/A
7225964	12/2006	Mastri et al.	N/A	N/A
7238195	12/2006	Viola	N/A	N/A
7246734	12/2006	Shelton, IV	N/A	N/A
7258262	12/2006	Mastri et al.	N/A	N/A
7267682	12/2006	Bender et al.	N/A	N/A
7278562	12/2006	Mastri et al.	N/A	N/A
7278563	12/2006	Green	N/A	N/A
7287682	12/2006	Ezzat et al.	N/A	N/A
7293685	12/2006	Ehrenfels et al.	N/A	N/A
7296722	12/2006	Ivanko	N/A	N/A
7296724	12/2006	Green et al.	N/A	N/A
7296772	12/2006	Wang	N/A	N/A

7300444	12/2006	Nielsen et al.	N/A	N/A
7303107	12/2006	Milliman et al.	N/A	N/A
7303108	12/2006	Shelton, IV	N/A	N/A
7308998	12/2006	Mastri et al.	N/A	N/A
7326232	12/2007	Viola et al.	N/A	N/A
7328828	12/2007	Ortiz et al.	N/A	N/A
7328829	12/2007	Arad et al.	N/A	N/A
7334717	12/2007	Rethy et al.	N/A	N/A
7354447	12/2007	Shelton, IV et al.	N/A	N/A
7357287	12/2007	Shelton, IV et al.	N/A	N/A
7364061	12/2007	Swayze et al.	N/A	N/A
7367485	12/2007	Shelton, IV et al.	N/A	N/A
7377928	12/2007	Zubik et al.	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7380696	12/2007	Shelton, IV et al.	N/A	N/A
7396356	12/2007	Mollenauer	N/A	N/A
7398907	12/2007	Racenet et al.	N/A	N/A
7399310	12/2007	Edoga et al.	N/A	N/A
7401720	12/2007	Durrani	N/A	N/A
7401721	12/2007	Holsten et al.	N/A	N/A
7404508	12/2007	Smith et al.	N/A	N/A
7404509	12/2007	Ortiz et al.	N/A	N/A
7407074	12/2007	Ortiz et al.	N/A	N/A
7407075	12/2007	Holsten et al.	N/A	N/A
7407077	12/2007	Ortiz et al.	N/A	N/A
7407078	12/2007	Shelton, IV et al.	N/A	N/A
7416101	12/2007	Shelton, IV et al.	N/A	N/A
7419080	12/2007	Smith et al.	N/A	N/A
7419081	12/2007	Ehrenfels et al.	N/A	N/A
7419495	12/2007	Menn et al.	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7424965	12/2007	Racenet et al.	N/A	N/A
7431189	12/2007	Shelton, IV et al.	N/A	N/A
7431730	12/2007	Viola	N/A	N/A
7434715	12/2007	Shelton, IV et al.	N/A	N/A
7434717	12/2007	Shelton, IV et al.	N/A	N/A
7438208	12/2007	Larson	N/A	N/A
7438209	12/2007	Hess et al.	N/A	N/A
7441684	12/2007	Shelton, IV et al.	N/A	N/A
7441685	12/2007	Boudreaux	N/A	N/A
7448525	12/2007	Shelton, IV et al.	N/A	N/A
7451904	12/2007	Shelton, IV	N/A	N/A
7455208	12/2007	Wales et al.	N/A	N/A
7455676	12/2007	Holsten et al.	N/A	N/A
7458494	12/2007	Matsutani et al.	N/A	N/A
7461767	12/2007	Viola et al.	N/A	N/A
7462185	12/2007	Knodel	N/A	N/A
7464846	12/2007	Shelton, IV et al.	N/A	N/A
7464847	12/2007	Viola et al.	N/A	N/A
7464848	12/2007	Green et al.	N/A	N/A

7464849	12/2007	Shelton, IV et al.	N/A	N/A
7467740	12/2007	Shelton, IV et al.	N/A	N/A
7472814	12/2008	Mastri et al.	N/A	N/A
7472815	12/2008	Shelton, IV et al.	N/A	N/A
7472816	12/2008	Holsten et al.	N/A	N/A
7473258	12/2008	Clauson et al.	N/A	N/A
7481347	12/2008	Roy	N/A	N/A
7481348	12/2008	Marczyk	N/A	N/A
7481349	12/2008	Holsten et al.	N/A	N/A
7481824	12/2008	Boudreaux et al.	N/A	N/A
7487899	12/2008	Shelton, IV et al.	N/A	N/A
7490749	12/2008	Schall et al.	N/A	N/A
7494039	12/2008	Racenet et al.	N/A	N/A
7500979	12/2008	Hueil et al.	N/A	N/A
7503474	12/2008	Hillstead et al.	N/A	N/A
7506790	12/2008	Shelton, IV	N/A	N/A
7506791	12/2008	Omaits et al.	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7513408	12/2008	Shelton, IV et al.	N/A	N/A
7517356	12/2008	Heinrich	N/A	N/A
7537602	12/2008	Whitman	N/A	N/A
7543729	12/2008	Ivanko	N/A	N/A
7543730	12/2008	Marczyk	N/A	N/A
7543731	12/2008	Green et al.	N/A	N/A
7552854	12/2008	Wixey et al.	N/A	N/A
7556185	12/2008	Viola	N/A	N/A
7556186	12/2008	Milliman	N/A	N/A
7559450	12/2008	Wales et al.	N/A	N/A
7559452	12/2008	Wales et al.	N/A	N/A
7559453	12/2008	Heinrich et al.	N/A	N/A
7559937	12/2008	de la Torre et al.	N/A	N/A
7565993	12/2008	Milliman et al.	N/A	N/A
7568603	12/2008	Shelton, IV et al.	N/A	N/A
7568604	12/2008	Ehrenfels et al.	N/A	N/A
7571845	12/2008	Viola	N/A	N/A
7575144	12/2008	Ortiz et al.	N/A	N/A
7584880	12/2008	Racenet et al.	N/A	N/A
7588174	12/2008	Holsten et al.	N/A	N/A
7588175	12/2008	Timm et al.	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7588177	12/2008	Racenet	N/A	N/A
7597229	12/2008	Boudreaux et al.	N/A	N/A
7597230	12/2008	Racenet et al.	N/A	N/A
7600663	12/2008	Green	N/A	N/A
7604150	12/2008	Boudreaux	N/A	N/A
7604151	12/2008	Hess et al.	N/A	N/A
7607557	12/2008	Shelton, IV et al.	N/A	N/A
7611038	12/2008	Racenet et al.	N/A	N/A
7617961	12/2008	Viola	N/A	N/A
7624902	12/2008	Marczyk et al.	N/A	N/A

7624903	12/2008	Green et al.	N/A	N/A
7631793	12/2008	Rethy et al.	N/A	N/A
7631794	12/2008	Rethy et al.	N/A	N/A
7635073	12/2008	Heinrich	N/A	N/A
7635074	12/2008	Olson et al.	N/A	N/A
7635373	12/2008	Ortiz	N/A	N/A
7637409	12/2008	Marczyk	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A
7641091	12/2009	Olson et al.	N/A	N/A
7641095	12/2009	Viola	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7648055	12/2009	Marczyk	N/A	N/A
7651017	12/2009	Ortiz et al.	N/A	N/A
7654431	12/2009	Hueil et al.	N/A	N/A
7658311	12/2009	Boudreaux	N/A	N/A
7658312	12/2009	Vidal et al.	N/A	N/A
7665646	12/2009	Prommersberger	N/A	N/A
7665647	12/2009	Shelton, IV et al.	N/A	N/A
7669746	12/2009	Shelton, IV	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7673780	12/2009	Shelton, IV et al.	N/A	N/A
7673781	12/2009	Swayze et al.	N/A	N/A
7673782	12/2009	Hess et al.	N/A	N/A
7673783	12/2009	Morgan et al.	N/A	N/A
7678121	12/2009	Knodel	N/A	N/A
7681772	12/2009	Green et al.	N/A	N/A
7682367	12/2009	Shah et al.	N/A	N/A
7682368	12/2009	Bombard et al.	N/A	N/A
7690547	12/2009	Racenet et al.	N/A	N/A
7694865	12/2009	Scirica	N/A	N/A
7699205	12/2009	Ivanko	N/A	N/A
7703653	12/2009	Shah et al.	N/A	N/A
7721931	12/2009	Shelton, IV et al.	N/A	N/A
7721933	12/2009	Ehrenfels et al.	N/A	N/A
7721935	12/2009	Racenet et al.	N/A	N/A
7726537	12/2009	Olson et al.	N/A	N/A
7726538	12/2009	Holsten et al.	N/A	N/A
7726539	12/2009	Holsten et al.	N/A	N/A
7731072	12/2009	Timm et al.	N/A	N/A
7735703	12/2009	Morgan et al.	N/A	N/A
7740159	12/2009	Shelton, IV et al.	N/A	N/A
7740160	12/2009	Viola	N/A	N/A
7743960	12/2009	Whitman et al.	N/A	N/A
7744628	12/2009	Viola	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7753248	12/2009	Viola	N/A	N/A
7757924	12/2009	Gerbi et al.	N/A	N/A
7757925	12/2009	Viola et al.	N/A	N/A
7762445	12/2009	Heinrich et al.	N/A	N/A
7766209	12/2009	Baxter, III et al.	N/A	N/A

7766210	12/2009	Shelton, IV et al.	N/A	N/A
7766924	12/2009	Bombard et al.	N/A	N/A
7766928	12/2009	Ezzat et al.	N/A	N/A
7770774	12/2009	Mastri et al.	N/A	N/A
7770775	12/2009	Shelton, IV et al.	N/A	N/A
7776060	12/2009	Mooradian et al.	N/A	N/A
7780055	12/2009	Scirica et al.	N/A	N/A
7784662	12/2009	Wales et al.	N/A	N/A
7789283	12/2009	Shah	N/A	N/A
7789889	12/2009	Zubik et al.	N/A	N/A
7793812	12/2009	Moore et al.	N/A	N/A
7793814	12/2009	Racenet et al.	N/A	N/A
7794475	12/2009	Hess et al.	N/A	N/A
7798385	12/2009	Boyden et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7799039	12/2009	Shelton, IV et al.	N/A	N/A
7810690	12/2009	Bilotti et al.	N/A	N/A
7810692	12/2009	Hall et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7815090	12/2009	Marczyk	N/A	N/A
7815091	12/2009	Marczyk	N/A	N/A
7815092	12/2009	Whitman et al.	N/A	N/A
7819296	12/2009	Hueil et al.	N/A	N/A
7819297	12/2009	Doll et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819299	12/2009	Shelton, IV et al.	N/A	N/A
7819896	12/2009	Racenet	N/A	N/A
7823760	12/2009	Zemlok et al.	N/A	N/A
7823761	12/2009	Boyden et al.	N/A	N/A
7824426	12/2009	Racenet et al.	N/A	N/A
7828186	12/2009	Wales	N/A	N/A
7828187	12/2009	Green et al.	N/A	N/A
7828188	12/2009	Jankowski	N/A	N/A
7828189	12/2009	Holsten et al.	N/A	N/A
7832408	12/2009	Shelton, IV et al.	N/A	N/A
7832611	12/2009	Boyden et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7834630	12/2009	Damadian et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837081	12/2009	Holsten et al.	N/A	N/A
7841503	12/2009	Sonnenschein et al.	N/A	N/A
7845533	12/2009	Marczyk et al.	N/A	N/A
7845534	12/2009	Viola et al.	N/A	N/A
7845535	12/2009	Scircia	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7845538	12/2009	Whitman	N/A	N/A
7850703	12/2009	Bombard et al.	N/A	N/A
7857183	12/2009	Shelton, IV	N/A	N/A
7857184	12/2009	Viola	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A

7857186	12/2009	Baxter, III et al.	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7861907	12/2010	Green et al.	N/A	N/A
7866524	12/2010	Krehel	N/A	N/A
7866525	12/2010	Scirica	N/A	N/A
7866526	12/2010	Green et al.	N/A	N/A
7866527	12/2010	Hall et al.	N/A	N/A
7866528	12/2010	Olson et al.	N/A	N/A
7870989	12/2010	Viola et al.	N/A	N/A
7886952	12/2010	Scirica et al.	N/A	N/A
7891532	12/2010	Mastri et al.	N/A	N/A
7891533	12/2010	Green et al.	N/A	N/A
7891534	12/2010	Wenchell et al.	N/A	N/A
7896214	12/2010	Farascioni	N/A	N/A
7900805	12/2010	Shelton, IV et al.	N/A	N/A
7901416	12/2010	Nolan et al.	N/A	N/A
7905380	12/2010	Shelton, IV et al.	N/A	N/A
7905381	12/2010	Baxter, III et al.	N/A	N/A
7909039	12/2010	Hur	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909221	12/2010	Viola et al.	N/A	N/A
7909224	12/2010	Prommersberger	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7913893	12/2010	Mastri et al.	N/A	N/A
7914543	12/2010	Roth et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7922061	12/2010	Shelton, IV et al.	N/A	N/A
7922063	12/2010	Zemlok et al.	N/A	N/A
7922064	12/2010	Boyden et al.	N/A	N/A
7926691	12/2010	Viola et al.	N/A	N/A
7926692	12/2010	Racenet et al.	N/A	N/A
7934628	12/2010	Wenchell et al.	N/A	N/A
7934630	12/2010	Shelton, IV et al.	N/A	N/A
7934631	12/2010	Balbierz et al.	N/A	N/A
7942300	12/2010	Rethy et al.	N/A	N/A
7942303	12/2010	Shah	N/A	N/A
7950560	12/2010	Zemlok et al.	N/A	N/A
7950561	12/2010	Aranyi	N/A	N/A
7950562	12/2010	Beardsley et al.	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7954683	12/2010	Knodel et al.	N/A	N/A
7954684	12/2010	Boudreaux	N/A	N/A
7954685	12/2010	Viola	N/A	N/A
7954686	12/2010	Baxter, III et al.	N/A	N/A
7954687	12/2010	Zemlok et al.	N/A	N/A
7959051	12/2010	Smith et al.	N/A	N/A
7963431	12/2010	Scirica	N/A	N/A
7963432	12/2010	Knodel et al.	N/A	N/A
7963433	12/2010	Whitman et al.	N/A	N/A
7967178	12/2010	Scirica et al.	N/A	N/A

7967179	12/2010	Olson et al.	N/A	N/A
7967180	12/2010	Scirica	N/A	N/A
7975894	12/2010	Boyden et al.	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7988026	12/2010	Knodel et al.	N/A	N/A
7988027	12/2010	Olson et al.	N/A	N/A
7988028	12/2010	Farascioni et al.	N/A	N/A
7992758	12/2010	Whitman et al.	N/A	N/A
7997468	12/2010	Farascioni	N/A	N/A
7997469	12/2010	Olson et al.	N/A	N/A
8002795	12/2010	Beetel	N/A	N/A
8006885	12/2010	Marczyk	N/A	N/A
8006887	12/2010	Marczyk	N/A	N/A
8007505	12/2010	Weller et al.	N/A	N/A
8007513	12/2010	Nalagatla et al.	N/A	N/A
8011550	12/2010	Aranyi et al.	N/A	N/A
8011551	12/2010	Marczyk et al.	N/A	N/A
8011552	12/2010	Ivanko	N/A	N/A
8011553	12/2010	Mastri et al.	N/A	N/A
8011555	12/2010	Tarinelli et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8015976	12/2010	Shah	N/A	N/A
8016177	12/2010	Bettuchi et al.	N/A	N/A
8016178	12/2010	Olson et al.	N/A	N/A
8020742	12/2010	Marczyk	N/A	N/A
8020743	12/2010	Shelton, IV	N/A	N/A
8028882	12/2010	Viola	N/A	N/A
8028883	12/2010	Stopek	N/A	N/A
8028884	12/2010	Sniffin et al.	N/A	N/A
8033438	12/2010	Scirica	N/A	N/A
8033440	12/2010	Wenchell et al.	N/A	N/A
8033441	12/2010	Marczyk	N/A	N/A
8033442	12/2010	Racenet et al.	N/A	N/A
8034077	12/2010	Smith et al.	N/A	N/A
8038044	12/2010	Viola	N/A	N/A
8038045	12/2010	Bettuchi et al.	N/A	N/A
8052024	12/2010	Viola et al.	N/A	N/A
8056787	12/2010	Boudreaux et al.	N/A	N/A
8056788	12/2010	Mastri et al.	N/A	N/A
8056791	12/2010	Whitman	N/A	N/A
8061577	12/2010	Racenet et al.	N/A	N/A
8066166	12/2010	Demmy et al.	N/A	N/A
8070033	12/2010	Milliman et al.	N/A	N/A
8070034	12/2010	Knodel	N/A	N/A
8070035	12/2010	Holsten et al.	N/A	N/A
8074858	12/2010	Marczyk	N/A	N/A
8074859	12/2010	Kostrzewski	N/A	N/A
8074862	12/2010	Shah	N/A	N/A
8083118	12/2010	Milliman et al.	N/A	N/A
8083119	12/2010	Prommersberger	N/A	N/A

8083120	12/2010	Shelton, IV et al.	N/A	N/A
8087563	12/2011	Milliman et al.	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091754	12/2011	Ehrenfels et al.	N/A	N/A
8091756	12/2011	Viola	N/A	N/A
8092493	12/2011	Marczyk	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8096460	12/2011	Blier et al.	N/A	N/A
8100309	12/2011	Marczyk	N/A	N/A
8100310	12/2011	Zemlok	N/A	N/A
8102008	12/2011	Wells	N/A	N/A
8113406	12/2011	Holsten et al.	N/A	N/A
8113407	12/2011	Holsten et al.	N/A	N/A
8113408	12/2011	Wenchell et al.	N/A	N/A
8113409	12/2011	Cohen et al.	N/A	N/A
8113410	12/2011	Hall et al.	N/A	N/A
8123101	12/2011	Racenet et al.	N/A	N/A
8127975	12/2011	Olson et al.	N/A	N/A
8127976	12/2011	Scirica et al.	N/A	N/A
8132703	12/2011	Milliman et al.	N/A	N/A
8132705	12/2011	Viola et al.	N/A	N/A
8132706	12/2011	Marczyk et al.	N/A	N/A
8136713	12/2011	Hathaway et al.	N/A	N/A
8141762	12/2011	Bedi et al.	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8157148	12/2011	Scirica	N/A	N/A
8157150	12/2011	Viola et al.	N/A	N/A
8157151	12/2011	Ingmanson et al.	N/A	N/A
8157152	12/2011	Holsten et al.	N/A	N/A
8162197	12/2011	Mastri et al.	N/A	N/A
8167185	12/2011	Shelton, IV et al.	N/A	N/A
8167186	12/2011	Racenet et al.	N/A	N/A
8172121	12/2011	Krehel	N/A	N/A
8172124	12/2011	Shelton, IV et al.	N/A	N/A
8181837	12/2011	Roy	N/A	N/A
8186555	12/2011	Shelton, IV et al.	N/A	N/A
8186557	12/2011	Cohen et al.	N/A	N/A
8186558	12/2011	Sapienza	N/A	N/A
8186559	12/2011	Whitman	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8193044	12/2011	Kenneth	N/A	N/A
8196795	12/2011	Moore et al.	N/A	N/A
8196796	12/2011	Shelton, IV et al.	N/A	N/A
8201721	12/2011	Zemlok et al.	N/A	N/A
8205619	12/2011	Shah et al.	N/A	N/A
8205780	12/2011	Sorrentino et al.	N/A	N/A
8205781	12/2011	Baxter, III et al.	N/A	N/A
8210412	12/2011	Marczyk	N/A	N/A
8210416	12/2011	Milliman et al.	N/A	N/A
8215532	12/2011	Marczyk	N/A	N/A

8216236	12/2011	Heinrich et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8220690	12/2011	Hess et al.	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A
8231040	12/2011	Zemlok et al.	N/A	N/A
8231041	12/2011	Marczyk et al.	N/A	N/A
8235272	12/2011	Nicholas et al.	N/A	N/A
8235273	12/2011	Olson et al.	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236010	12/2011	Ortiz et al.	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8245897	12/2011	Tzakis et al.	N/A	N/A
8245898	12/2011	Smith et al.	N/A	N/A
8245899	12/2011	Swensgard et al.	N/A	N/A
8245931	12/2011	Shigeta	N/A	N/A
8252009	12/2011	Weller et al.	N/A	N/A
8256653	12/2011	Farascioni	N/A	N/A
8256654	12/2011	Bettuchi et al.	N/A	N/A
8256655	12/2011	Sniffin et al.	N/A	N/A
8256656	12/2011	Milliman et al.	N/A	N/A
8267300	12/2011	Boudreaux	N/A	N/A
8272551	12/2011	Knodel et al.	N/A	N/A
8272553	12/2011	Mastri et al.	N/A	N/A
8272554	12/2011	Whitman et al.	N/A	N/A
8276594	12/2011	Shah	N/A	N/A
8276801	12/2011	Zemlok et al.	N/A	N/A
8281973	12/2011	Wenchell et al.	N/A	N/A
8286847	12/2011	Taylor	N/A	N/A
8286848	12/2011	Wenchell et al.	N/A	N/A
8286850	12/2011	Viola	N/A	N/A
8292146	12/2011	Holsten et al.	N/A	N/A
8292147	12/2011	Viola	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292149	12/2011	Ivanko	N/A	N/A
8292150	12/2011	Bryant	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman et al.	N/A	N/A
8292153	12/2011	Jankowski	N/A	N/A
8292154	12/2011	Marczyk	N/A	N/A
8292155	12/2011	Shelton, IV et al.	N/A	N/A
8292156	12/2011	Kostrzewski	N/A	N/A
8292158	12/2011	Sapienza	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8308041	12/2011	Kostrzewski	N/A	N/A
8308042	12/2011	Aranyi	N/A	N/A
8308043	12/2011	Bindra et al.	N/A	N/A
8308044	12/2011	Viola	N/A	N/A
8308046	12/2011	Prommersberger	N/A	N/A

8308757	12/2011	Hillstead et al.	N/A	N/A
8317070	12/2011	Hueil et al.	N/A	N/A
8317071	12/2011	Knodel	N/A	N/A
8322455	12/2011	Shelton, IV et al.	N/A	N/A
8322589	12/2011	Boudreaux	N/A	N/A
8328061	12/2011	Kasvikis	N/A	N/A
8328065	12/2011	Shah	N/A	N/A
8333313	12/2011	Boudreaux et al.	N/A	N/A
8336751	12/2011	Scirica	N/A	N/A
8336753	12/2011	Olson et al.	N/A	N/A
8336754	12/2011	Cappola et al.	N/A	N/A
8342377	12/2012	Milliman et al.	N/A	N/A
8342378	12/2012	Marczyk et al.	N/A	N/A
8342379	12/2012	Whitman et al.	N/A	N/A
8342380	12/2012	Viola	N/A	N/A
8348123	12/2012	Scirica et al.	N/A	N/A
8348124	12/2012	Scirica	N/A	N/A
8348125	12/2012	Viola et al.	N/A	N/A
8348126	12/2012	Olson et al.	N/A	N/A
8348127	12/2012	Marczyk	N/A	N/A
8348129	12/2012	Bedi et al.	N/A	N/A
8348130	12/2012	Shah et al.	N/A	N/A
8348131	12/2012	Omaits et al.	N/A	N/A
8353437	12/2012	Boudreaux	N/A	N/A
8353440	12/2012	Whitman et al.	N/A	N/A
8356740	12/2012	Knodel	N/A	N/A
8357174	12/2012	Roth et al.	N/A	N/A
8360294	12/2012	Scirica	N/A	N/A
8360297	12/2012	Shelton, IV et al.	N/A	N/A
8360298	12/2012	Farascioni et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8365971	12/2012	Knodel	N/A	N/A
8365972	12/2012	Aranyi et al.	N/A	N/A
8365973	12/2012	White et al.	N/A	N/A
8365976	12/2012	Hess et al.	N/A	N/A
8371491	12/2012	Huitema et al.	N/A	N/A
8371492	12/2012	Aranyi et al.	N/A	N/A
8371493	12/2012	Aranyi et al.	N/A	N/A
8381828	12/2012	Whitman et al.	N/A	N/A
8381961	12/2012	Holsten et al.	N/A	N/A
8387848	12/2012	Johnson et al.	N/A	N/A
8387849	12/2012	Buesseler et al.	N/A	N/A
8387850	12/2012	Hathaway et al.	N/A	N/A
8388652	12/2012	Viola	N/A	N/A
8393513	12/2012	Jankowski	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8393516	12/2012	Kostrzewski	N/A	N/A
8397971	12/2012	Yates et al.	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8403195	12/2012	Beardsley et al.	N/A	N/A

8403196	12/2012	Beardsley et al.	N/A	N/A
8403197	12/2012	Vidal et al.	N/A	N/A
8403198	12/2012	Sorrentino et al.	N/A	N/A
8403956	12/2012	Thompson et al.	N/A	N/A
8408439	12/2012	Huang et al.	N/A	N/A
8408440	12/2012	Olson et al.	N/A	N/A
8408442	12/2012	Racenet et al.	N/A	N/A
8413868	12/2012	Cappola	N/A	N/A
8413869	12/2012	Heinrich	N/A	N/A
8413871	12/2012	Racenet et al.	N/A	N/A
8418904	12/2012	Wenchell et al.	N/A	N/A
8418905	12/2012	Milliman	N/A	N/A
8418906	12/2012	Farascioni et al.	N/A	N/A
8418907	12/2012	Johnson et al.	N/A	N/A
8418908	12/2012	Beardsley	N/A	N/A
8419768	12/2012	Marczyk	N/A	N/A
8424735	12/2012	Viola et al.	N/A	N/A
8424736	12/2012	Scirica et al.	N/A	N/A
8424737	12/2012	Scirica	N/A	N/A
8424739	12/2012	Racenet et al.	N/A	N/A
8424740	12/2012	Shelton, IV et al.	N/A	N/A
8439244	12/2012	Holcomb et al.	N/A	N/A
8439245	12/2012	Knodel et al.	N/A	N/A
8439246	12/2012	Knodel	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8444037	12/2012	Nicholas et al.	N/A	N/A
8444038	12/2012	Farascioni et al.	N/A	N/A
8448832	12/2012	Viola et al.	N/A	N/A
8453652	12/2012	Stopek	N/A	N/A
8453905	12/2012	Holcomb et al.	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8453907	12/2012	Laurent et al.	N/A	N/A
8453908	12/2012	Bedi et al.	N/A	N/A
8453909	12/2012	Olson et al.	N/A	N/A
8453910	12/2012	Bettuchi et al.	N/A	N/A
8453912	12/2012	Mastri et al.	N/A	N/A
8453913	12/2012	Milliman	N/A	N/A
8453914	12/2012	Laurent et al.	N/A	N/A
8454628	12/2012	Smith et al.	N/A	N/A
8459520	12/2012	Giordano et al.	N/A	N/A
8459521	12/2012	Zemlok et al.	N/A	N/A
8459522	12/2012	Marczyk	N/A	N/A
8459523	12/2012	Whitman	N/A	N/A
8459524	12/2012	Pribanic et al.	N/A	N/A
8459525	12/2012	Yates et al.	N/A	N/A
8464922	12/2012	Marczyk	N/A	N/A
8464923	12/2012	Shelton, IV	N/A	N/A
8469252	12/2012	Holcomb et al.	N/A	N/A
8469254	12/2012	Czernik et al.	N/A	N/A
8474677	12/2012	Woodard, Jr. et al.	N/A	N/A

8479967	12/2012	Marczyk	N/A	N/A
8479968	12/2012	Hodgkinson et al.	N/A	N/A
8479969	12/2012	Shelton, IV	N/A	N/A
8485412	12/2012	Shelton, IV et al.	N/A	N/A
8490852	12/2012	Viola	N/A	N/A
8496152	12/2012	Viola	N/A	N/A
8496154	12/2012	Marczyk et al.	N/A	N/A
8496156	12/2012	Sniffin et al.	N/A	N/A
8496683	12/2012	Prommersberger et al.	N/A	N/A
8499993	12/2012	Shelton, IV et al.	N/A	N/A
8505799	12/2012	Viola et al.	N/A	N/A
8505802	12/2012	Viola et al.	N/A	N/A
8511575	12/2012	Cok	N/A	N/A
8512359	12/2012	Whitman et al.	N/A	N/A
8512402	12/2012	Marczyk et al.	N/A	N/A
8517240	12/2012	Mata et al.	N/A	N/A
8517241	12/2012	Nicholas et al.	N/A	N/A
8517243	12/2012	Giordano et al.	N/A	N/A
8517244	12/2012	Shelton, IV et al.	N/A	N/A
8523041	12/2012	Ishitsuki et al.	N/A	N/A
8523042	12/2012	Masiakos et al.	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8534528	12/2012	Shelton, IV	N/A	N/A
8540128	12/2012	Shelton, IV et al.	N/A	N/A
8540129	12/2012	Baxter, III et al.	N/A	N/A
8540130	12/2012	Moore et al.	N/A	N/A
8540131	12/2012	Swayze	N/A	N/A
8540733	12/2012	Whitman et al.	N/A	N/A
8544711	12/2012	Ma et al.	N/A	N/A
8550325	12/2012	Cohen et al.	N/A	N/A
8556151	12/2012	Viola	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8561873	12/2012	Ingmanson et al.	N/A	N/A
8561874	12/2012	Scirica	N/A	N/A
8567656	12/2012	Shelton, IV et al.	N/A	N/A
8573461	12/2012	Shelton, IV et al.	N/A	N/A
8573463	12/2012	Scirica et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579177	12/2012	Beetel	N/A	N/A
8584919	12/2012	Hueil et al.	N/A	N/A
8584920	12/2012	Hodgkinson	N/A	N/A
8590762	12/2012	Hess et al.	N/A	N/A
8596515	12/2012	Okoniewski	N/A	N/A
8602288	12/2012	Shelton, IV et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8608046	12/2012	Laurent et al.	N/A	N/A
8608047	12/2012	Holsten et al.	N/A	N/A
8613383	12/2012	Beckman et al.	N/A	N/A

8613384	12/2012	Pastorelli et al.	N/A	N/A
8616427	12/2012	Viola	N/A	N/A
8616430	12/2012	Stopek et al.	N/A	N/A
8627994	12/2013	Zemlok et al.	N/A	N/A
8628544	12/2013	Farascioni	N/A	N/A
8631988	12/2013	Viola	N/A	N/A
8631989	12/2013	Aranyi et al.	N/A	N/A
8631991	12/2013	Cropper et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8632535	12/2013	Shelton, IV et al.	N/A	N/A
8636187	12/2013	Hueil et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636192	12/2013	Farascioni et al.	N/A	N/A
8636762	12/2013	Whitman et al.	N/A	N/A
8636766	12/2013	Milliman et al.	N/A	N/A
8640940	12/2013	Ohdaira	N/A	N/A
8657174	12/2013	Yates et al.	N/A	N/A
8657177	12/2013	Scirica et al.	N/A	N/A
8657178	12/2013	Hueil et al.	N/A	N/A
8662371	12/2013	Viola	N/A	N/A
8668129	12/2013	Olson	N/A	N/A
8672206	12/2013	Aranyi et al.	N/A	N/A
8672208	12/2013	Hess et al.	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
8678263	12/2013	Viola	N/A	N/A
8678990	12/2013	Nazer et al.	N/A	N/A
8679155	12/2013	Knodel et al.	N/A	N/A
8684247	12/2013	Scirica et al.	N/A	N/A
8684249	12/2013	Racenet et al.	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8690039	12/2013	Beardsley et al.	N/A	N/A
8695865	12/2013	Smith et al.	N/A	N/A
8695866	12/2013	Leimbach et al.	N/A	N/A
8701958	12/2013	Shelton, IV et al.	N/A	N/A
8701959	12/2013	Shah	N/A	N/A
8701961	12/2013	Ivanko	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8714429	12/2013	Demmy	N/A	N/A
8715277	12/2013	Weizman	N/A	N/A
8720766	12/2013	Hess et al.	N/A	N/A
8721630	12/2013	Ortiz et al.	N/A	N/A
8727197	12/2013	Hess et al.	N/A	N/A
8727200	12/2013	Roy	N/A	N/A
8733612	12/2013	Ma	N/A	N/A
8740034	12/2013	Morgan et al.	N/A	N/A
8740039	12/2013	Farascioni	N/A	N/A
8746529	12/2013	Shelton, IV et al.	N/A	N/A
8746530	12/2013	Giordano et al.	N/A	N/A
8746535	12/2013	Shelton, IV et al.	N/A	N/A
8752748	12/2013	Whitman et al.	N/A	N/A

8752749	12/2013	Moore et al.	N/A	N/A
8757465	12/2013	Woodard, Jr. et al.	N/A	N/A
8758391	12/2013	Swayze et al.	N/A	N/A
8763877	12/2013	Schall et al.	N/A	N/A
8763879	12/2013	Shelton, IV et al.	N/A	N/A
8770458	12/2013	Scirica	N/A	N/A
8777082	12/2013	Scirica	N/A	N/A
8783541	12/2013	Shelton, IV et al.	N/A	N/A
8783542	12/2013	Riestenberg et al.	N/A	N/A
8789737	12/2013	Hodgkinson et al.	N/A	N/A
8789738	12/2013	Knodel et al.	N/A	N/A
8789739	12/2013	Swensgard	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8800840	12/2013	Jankowski	N/A	N/A
8800841	12/2013	Ellerhorst et al.	N/A	N/A
8808311	12/2013	Heinrich et al.	N/A	N/A
8814024	12/2013	Woodard, Jr. et al.	N/A	N/A
8814025	12/2013	Miller et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8820607	12/2013	Marczyk	N/A	N/A
8827133	12/2013	Shelton, IV et al.	N/A	N/A
8827134	12/2013	Viola et al.	N/A	N/A
8833631	12/2013	Munro, III et al.	N/A	N/A
8833632	12/2013	Swensgard	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
8840603	12/2013	Shelton, IV et al.	N/A	N/A
8844788	12/2013	Knodel	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8851355	12/2013	Aranyi et al.	N/A	N/A
8857693	12/2013	Schuckmann et al.	N/A	N/A
8864007	12/2013	Widenhouse et al.	N/A	N/A
8864009	12/2013	Shelton, IV et al.	N/A	N/A
8875971	12/2013	Hall et al.	N/A	N/A
8875972	12/2013	Weisenburgh, II et al.	N/A	N/A
8893949	12/2013	Shelton, IV et al.	N/A	N/A
8893950	12/2013	Marczyk	N/A	N/A
8899461	12/2013	Farascioni	N/A	N/A
8899463	12/2013	Schall et al.	N/A	N/A
8899464	12/2013	Hueil et al.	N/A	N/A
8900616	12/2013	Belcheva et al.	N/A	N/A
8920435	12/2013	Smith et al.	N/A	N/A
8925782	12/2014	Shelton, IV	N/A	N/A
8926598	12/2014	Mollere et al.	N/A	N/A
8931682	12/2014	Timm et al.	N/A	N/A
8931693	12/2014	Kumar et al.	N/A	N/A
8955732	12/2014	Zemlok et al.	N/A	N/A
8958429	12/2014	Shukla et al.	N/A	N/A
8960517	12/2014	Lee	N/A	N/A

8967443	12/2014	McCuen	N/A	N/A
8967446	12/2014	Beardsley et al.	N/A	N/A
8973803	12/2014	Hall et al.	N/A	N/A
8978954	12/2014	Shelton, IV et al.	N/A	N/A
8978956	12/2014	Schall et al.	N/A	N/A
8998060	12/2014	Bruewer et al.	N/A	N/A
9005230	12/2014	Yates et al.	N/A	N/A
9010607	12/2014	Kostrzewski	N/A	N/A
9016539	12/2014	Kostrzewski et al.	N/A	N/A
9016541	12/2014	Viola et al.	N/A	N/A
9016542	12/2014	Shelton, IV et al.	N/A	N/A
9016546	12/2014	Demmy et al.	N/A	N/A
9022271	12/2014	Scirica	N/A	N/A
9027817	12/2014	Milliman et al.	N/A	N/A
9033203	12/2014	Woodard, Jr. et al.	N/A	N/A
9044228	12/2014	Woodard, Jr. et al.	N/A	N/A
9044229	12/2014	Scheib et al.	N/A	N/A
9050084	12/2014	Schmid et al.	N/A	N/A
9055941	12/2014	Schmid et al.	N/A	N/A
9055943	12/2014	Zemlok et al.	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9089326	12/2014	Krumanaker et al.	N/A	N/A
9101359	12/2014	Smith et al.	N/A	N/A
9107663	12/2014	Swensgard	N/A	N/A
9107664	12/2014	Marczyk	N/A	N/A
9113862	12/2014	Morgan et al.	N/A	N/A
9113864	12/2014	Morgan et al.	N/A	N/A
9113870	12/2014	Viola	N/A	N/A
9113872	12/2014	Viola	N/A	N/A
9113880	12/2014	Zemlok et al.	N/A	N/A
9125649	12/2014	Bruewer et al.	N/A	N/A
9138225	12/2014	Huang et al.	N/A	N/A
9155537	12/2014	Katre et al.	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9192378	12/2014	Aranyi et al.	N/A	N/A
9192379	12/2014	Aranyi et al.	N/A	N/A
9192384	12/2014	Bettuchi	N/A	N/A
9198644	12/2014	Balek et al.	N/A	N/A
9198661	12/2014	Swensgard	N/A	N/A
9204876	12/2014	Cappola et al.	N/A	N/A
9216019	12/2014	Schmid et al.	N/A	N/A
9216020	12/2014	Zhang et al.	N/A	N/A
9220500	12/2014	Swayze et al.	N/A	N/A
9220501	12/2014	Baxter, III et al.	N/A	N/A
9220502	12/2014	Zemlok et al.	N/A	N/A
9232941	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9232944	12/2015	Cappola et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A

9254180	12/2015	Huitema et al.	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9271728	12/2015	Gupta et al.	N/A	N/A
9277919	12/2015	Timmer et al.	N/A	N/A
9282962	12/2015	Schmid et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9289209	12/2015	Gurumurthy et al.	N/A	N/A
9289210	12/2015	Baxter, III et al.	N/A	N/A
9289225	12/2015	Shelton, IV et al.	N/A	N/A
9295464	12/2015	Shelton, IV et al.	N/A	N/A
9295465	12/2015	Farascioni	N/A	N/A
9301752	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9301753	12/2015	Aldridge et al.	N/A	N/A
9301757	12/2015	Williams	N/A	N/A
9307965	12/2015	Ming et al.	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A
9307989	12/2015	Shelton, IV et al.	N/A	N/A
9314246	12/2015	Shelton, IV et al.	N/A	N/A
9320518	12/2015	Henderson et al.	N/A	N/A
9320521	12/2015	Shelton, IV et al.	N/A	N/A
9326767	12/2015	Koch, Jr. et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9345477	12/2015	Anim et al.	N/A	N/A
9345478	12/2015	Knodel	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345780	12/2015	Manoharan et al.	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9351732	12/2015	Hodgkinson	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9364217	12/2015	Kostrzewski et al.	N/A	N/A
9364218	12/2015	Scirica	N/A	N/A
9364219	12/2015	Olson et al.	N/A	N/A
9364220	12/2015	Williams	N/A	N/A
9364233	12/2015	Alexander, III et al.	N/A	N/A
9370358	12/2015	Shelton, IV et al.	N/A	N/A
9370362	12/2015	Petty et al.	N/A	N/A
9386983	12/2015	Swensgard et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9386988	12/2015	Baxter, III et al.	N/A	N/A
9393018	12/2015	Wang et al.	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
9402604	12/2015	Williams et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9433419	12/2015	Gonzalez et al.	N/A	N/A
9433420	12/2015	Hodgkinson	N/A	N/A
9445810	12/2015	Cappola	N/A	N/A
9445813	12/2015	Shelton, IV et al.	N/A	N/A
9451959	12/2015	Patankar et al.	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A

9468439	12/2015	Cappola et al.	N/A	N/A
9480476	12/2015	Aldridge et al.	N/A	N/A
9480492	12/2015	Aranyi et al.	N/A	N/A
9492171	12/2015	Patenaude	N/A	N/A
9498212	12/2015	Racenet et al.	N/A	N/A
9510827	12/2015	Kostrzewski	N/A	N/A
9517065	12/2015	Simms et al.	N/A	N/A
9517066	12/2015	Racenet et al.	N/A	N/A
9539007	12/2016	Dhakad et al.	N/A	N/A
9549735	12/2016	Shelton, IV et al.	N/A	N/A
9622746	12/2016	Simms	N/A	A61B 17/07207
2004/0108357	12/2003	Milliman et al.	N/A	N/A
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2010/0069942	12/2009	Shelton, IV	N/A	N/A
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2010/0127041	12/2009	Morgan et al.	N/A	N/A
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2010/0147922	12/2009	Olson	N/A	N/A
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2011/0163149	12/2010	Viola	N/A	N/A
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2012/0187179	12/2011	Gleiman	N/A	N/A
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2012/0234897	12/2011	Shelton, IV et al.	N/A	N/A
2012/0241492	12/2011	Shelton, IV et al.	N/A	N/A
2012/0241493	12/2011	Baxter, III et al.	N/A	N/A
2012/0241504	12/2011	Soltz et al.	N/A	N/A
2012/0248169	12/2011	Widenhouse et al.	N/A	N/A
2012/0286021	12/2011	Kostrzewski	N/A	N/A
2012/0286022	12/2011	Olson et al.	N/A	N/A
2012/0298722	12/2011	Iess et al.	N/A	N/A
2013/0008937	12/2012	Viola	N/A	N/A
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2013/0068816	12/2012	Mandakolathur Vasudevan et al.	N/A	N/A
2013/0068818	12/2012	Kasvikis	N/A	N/A
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2013/0098964	12/2012	Smith et al.	N/A	N/A
2013/0098966	12/2012	Kostrzewski et al.	N/A	N/A
2013/0098970	12/2012	Racenet et al.	N/A	N/A
2013/0105545	12/2012	Burbank	N/A	N/A
2013/0105548	12/2012	Hodgkinson et al.	N/A	N/A
2013/0105552	12/2012	Weir et al.	N/A	N/A
2013/0105553	12/2012	Racenet et al.	N/A	N/A
2013/0112730	12/2012	Whitman et al.	N/A	N/A
2013/0119109	12/2012	Farascioni et al.	N/A	N/A
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2013/0146642	12/2012	Shelton, IV et al.	N/A	N/A
2013/0153636	12/2012	Shelton, IV et al.	N/A	N/A
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2014/0076955	12/2013	Lorenz	N/A	N/A
2014/0131419	12/2013	Bettuchi	N/A	N/A
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2014/0166726	12/2013	Schellin et al.	N/A	N/A
2014/0175146	12/2013	Knodel	N/A	N/A
2014/0175150	12/2013	Shelton, IV et al.	N/A	N/A
2014/0200596	12/2013	Weir	606/142	A61B 50/13
2014/0203062	12/2013	Viola	N/A	N/A
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2014/0239043	12/2013	Simms et al.	N/A	N/A
2014/0239044	12/2013	Hoffman	N/A	N/A
2014/0239047	12/2013	Hodgkinson et al.	N/A	N/A
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2014/0246475	12/2013	Hall et al.	N/A	N/A
2014/0246478	12/2013	Baber et al.	N/A	N/A
2014/0252062	12/2013	Mozdzierz	N/A	N/A
2014/0252064	12/2013	Mozdzierz et al.	N/A	N/A
2014/0252065	12/2013	Hessler et al.	N/A	N/A
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2014/0263544	12/2013	Ranucci et al.	N/A	N/A
2014/0263546	12/2013	Aranyi	N/A	N/A
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2014/0263552	12/2013	Hall et al.	N/A	N/A
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2014/0263557	12/2013	Schaller	N/A	N/A
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2015/0053744	12/2014	Swayze et al.	N/A	N/A
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Primary Examiner: Long; Robert F

Background/Summary

FIELD

(1) The disclosure is directed to surgical stapling devices and, more particularly, to endoscopic surgical stapling devices.

BACKGROUND

(2) Surgical stapling devices for performing surgical procedures endoscopically are well known. Such devices are available in a variety of different configurations, e.g., linear, curved, circular, etc., and are suitable for use in a variety of different procedures. Linear surgical stapling devices include a tool assembly having an anvil and a staple cartridge that are pivotably coupled to each other at their proximal ends between open and clamped positions. In order to better navigate the tool assembly to a surgical site endoscopically, the tool assembly may include a dissector tip that

extends from a distal end of the tool assembly. Typically, the dissector tip is supported on the tool assembly and is configured to separate target tissue from body tissue to facilitate placement of the tool assembly about the target tissue. During some surgical procedures, it is desirable to have an angled dissector tip whereas in other surgical procedures it is desirable to have a linear dissector tip.

(3) A continuing need exists in the art for a surgical stapling device that is better suited to access to a variety of surgical sites.

SUMMARY

(4) One aspect of this disclosure is directed a surgical stapler including an elongate body and a tool assembly. The elongate body has a proximal portion and a distal portion. The tool assembly defines a longitudinal axis and includes a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly has a proximal portion and a distal portion and is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The dissector tip includes a body having a proximal end and a distal end. The body has a thickness that decreases towards the distal end. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis.

(5) In aspects of the disclosure, the dissector tip includes a tip beam having a proximal portion and a deformable distal beam portion, wherein the deformable distal beam portion is bendable between the first and second configurations.

(6) In some aspects of the disclosure, the distal portion of the anvil assembly includes a bracket and the proximal portion of the tip beam is secured to the bracket.

(7) In certain aspects of the disclosure, the proximal portion of the tip beam is secured to the bracket of the anvil assembly by welding or other means (mechanical snap, adhesive, pin, etc).

(8) In aspects of the disclosure, the dissector tip includes a tip beam cover that is supported on the deformable distal beam portion and has an atraumatic configuration.

(9) In some aspects of the disclosure, the tip beam cover is formed of a biocompatible material selected from the group consisting of polymeric materials, metals, ceramics, and elastomeric materials.

(10) In certain aspects of the disclosure, the dissector tip includes a body having a proximal portion and a distal portion, wherein the proximal portion includes a hinge portion that is pivotably coupled to the distal portion of the anvil assembly.

(11) In aspects of the disclosure, the distal portion of the anvil assembly includes a clevis and the hinge portion is coupled to the clevis by a clevis pin.

(12) In some aspects of the disclosure, the tool assembly includes retaining structure configured to retain the dissector tip in one of the first and second configurations.

(13) In certain aspects of the disclosure, the retaining structure includes a protrusion formed on one of the distal portion of the anvil assembly or the proximal portion of the dissector tip and a plurality of recesses formed on the other of the distal portion of the anvil assembly or the proximal portion of the dissector tip.

(14) In one aspect of the disclosure, the dissector tip has a width that decreases towards the distal end of the dissector tip.

(15) Another aspect of the disclosure is directed to a tool assembly including a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly includes a proximal portion and a distal portion and is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The tool assembly defines a longitudinal axis. The dissector tip includes a body having a proximal end and a distal end and has a thickness that decreases towards the distal end. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis.

(16) Another aspect of the disclosure is directed to a tool assembly defining a longitudinal axis and including a cartridge assembly, an anvil assembly, and a dissector tip. The anvil assembly includes a base portion and an anvil portion. The base portion is coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position. The anvil portion is secured to the base portion and includes a plurality of staple deforming pockets. The anvil portion includes a distal portion supporting a bracket. The dissector tip includes a body having a tapered distal surface and a thickness that decreases in the distal direction. The body is movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis. The body of the dissector tip includes a tip beam having a proximal portion and a deformable distal beam portion. The proximal portion of the dissector tip is secured to the bracket of the anvil assembly and the deformable distal beam portion is bendable between the first and second configurations.

(17) Other features of the disclosure will be appreciated from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of the disclosed surgical stapling device are described herein below with reference to the drawings, wherein:
- (2) FIG. 1 is a side perspective view of exemplary aspects of the disclosed surgical stapling device including a reload assembly with a universal tip;
- (3) FIG. 2 is an enlarged view of the indicated area of detail shown in FIG. 1;
- (4) FIG. 3 is a perspective bottom view of an anvil assembly of the reload assembly of the surgical stapling device shown in FIG. 1;
- (5) FIG. 4 is an exploded, perspective view of the anvil assembly shown in FIG. 3;
- (6) FIG. 5 is an enlarged cutaway, perspective view of a distal portion of an anvil body and tip beam of the anvil assembly shown in FIG. 4;
- (7) FIG. 6 is a side perspective view of a tool assembly of the reload assembly shown in FIG. 1 in a clamped position;
- (8) FIG. 7 is a cross-sectional view taken along section line 7-7 of FIG. 6 with the anvil tip in a first position;
- (9) FIG. 8 is a cross-sectional view taken along section line 7-7 of FIG. 6 with the anvil tip in a second position;
- (10) FIG. 9 is a cross-sectional view taken along section line 7-7 of FIG. 6 with the anvil tip in a third position;
- (11) FIG. 10 is a side view of other exemplary aspects of a tool assembly of the reload assembly of the stapling device shown in FIG. 1 with the tool assembly in the clamped position;
- (12) FIG. 11 is an enlarged side perspective view of a distal end of the tool assembly shown in FIG. 10 with the tool assembly in the clamped position;
- (13) FIG. 12 is a side perspective, exploded view of the anvil assembly of the tool assembly shown in FIG. 10;
- (14) FIG. 13 is an enlarged view of the distal end of the anvil base of the anvil assembly shown in FIG. 10;
- (15) FIG. 14 is a perspective view from the proximal end of the dissector tip of the anvil assembly shown in FIG. 11;
- (16) FIG. 15 is a cross-sectional view taken along section line 15-15 of FIG. 11 with the dissector tip in a first position; and
- (17) FIG. 16 is a cross-sectional view taken along section line 15-15 of FIG. 11 with the dissector

tip in a second position.

DETAILED DESCRIPTION

(18) The disclosed surgical stapling device will now be described in detail with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. However, it is to be understood that the aspects of the disclosure described herein are merely exemplary of the disclosure and may be embodied in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosure in virtually any appropriately detailed structure.

(19) In this description, the term “proximal” is used generally to refer to that portion of the device that is closer to a clinician, while the term “distal” is used generally to refer to that portion of the device that is farther from the clinician. In addition, the term “endoscopic” is used generally to refer to endoscopic, laparoscopic, arthroscopic, and/or any other procedure conducted through a small diameter incision or cannula. Further, the term “clinician” is used generally to refer to medical personnel including doctors, nurses, and support personnel. As used herein, the terms “parallel” and “aligned” are understood to include relative configurations that are substantially parallel, and substantially aligned, i.e., up to about + or -10 degrees from true parallel or true alignment.

(20) The disclosed surgical stapling device includes a tool assembly that supports a dissector tip that is adjustable to allow a clinician to position the dissector tip in a configuration that facilitates easy access to a surgical site. Some aspects of the disclosure are directed to a dissector tip that is bendable between a first orientation in which an axis defined by dissector tip is parallel to a longitudinal axis of the tool assembly and a second position in which the axis defined by the dissector tip defines an acute angle with the longitudinal axis of the tool assembly. Other aspects of the disclosure are directed to a dissector tip that is pivotable between a first orientation in which the axis defined by the dissector tip is parallel to a longitudinal axis of the tool assembly and a second position in which the axis defined by the dissector tip defines an acute angle with the longitudinal axis of the tool assembly.

(21) FIG. 1 illustrates the disclosed surgical stapling device **10** including a handle assembly **12**, an elongate body **14**, and a tool assembly **100**. Although not described in detail herein, the tool assembly **100** can form part of a reload assembly **16** that is releasably coupled to the elongate body **14** and includes a proximal body portion **16a** and the tool assembly **100**. Alternately, the tool assembly **100** can be fixedly secured to a distal portion of the elongate body **14**. The handle assembly **12** includes a hand grip **18**, a plurality of actuator buttons **20**, and a rotation knob **22**. The rotation knob **22** facilitates rotation of the elongate body **14** and the tool assembly **100** in relation to the handle assembly **12**. The actuator buttons **20** control operation of the various functions of the stapling device **10** including approximation, firing and cutting. Although the stapling device **10** is illustrated as an electrically powered stapling device, it is envisioned that the disclosed tool assembly **100** would also be suitable for use with a manually powered surgical stapling device. U.S. Pat. No. 9,055,943 (the '943 Patent) discloses a surgical stapling device including a powered handle assembly and U.S. Pat. No. 6,241,139 (the '139 Patent) discloses a manually actuated handle assembly. The tool assembly **100** includes a cartridge assembly **112**, an anvil assembly **114**, and a dissector tip **120**. For a more detailed description of the cartridge assembly **112** of the tool assembly **100**, see, e.g., the '943 and '139 Patents.

(22) FIGS. 2-7 illustrate the anvil assembly **114** of the tool assembly **100** which defines a longitudinal axis “X” and includes a base portion **116** and an anvil portion **118**. The base portion **116** includes a proximal mounting portion **116a** and a longitudinal rib **116b** that extends distally from the proximal mounting portion **116a**. The proximal mounting portion **116a** includes spaced

through bores **116c** that receive pivot members **117** (FIG. 1) that pivotably couple the cartridge assembly **112** to the anvil assembly **114**. The anvil portion **118** is secured to the base portion **116** by, e.g., welding, and includes an upper surface **122**, a proximal portion **118a**, and a distal portion **118b**. The upper surface **122** of the anvil portion **118** defines a plurality of rows of staple deforming pockets **126** that are positioned on opposite sides of a knife slot **128**. The distal portion **118b** of the anvil portion **118** includes a support plate or bracket **130**.

(23) The dissector tip **120** includes a tip beam **140** and a tip beam cover **142**. The tip beam **140** includes a body having a proximal portion **144** and a deformable distal beam portion **146**. The tip beam **140** is secured to the support bracket **130** on the distal portion **118b** of the anvil portion **118** and is formed from a substantially rigid material that can be bent and shaped to a desired configuration such as a metal. The tip beam cover **142** has an atraumatic configuration and is formed of any suitable biocompatible material including polymeric materials, metals, ceramics, elastomeric materials, etc. In certain aspects of the disclosure, the tip beam cover **142** has rounded edges **156** and a tapered distal tip **158** (FIG. 4) that decreases in width in the distal direction. The tip beam cover **142** also has a thickness that decreases in the distal direction such that the cover **142** is thinnest at its distal end. The reduced thickness allows for the dissector tip **120** to more easily navigate through and dissect tissue. Other tip beam cover configurations are envisioned.

(24) FIGS. 8 and 9 illustrate the distal portion of the tool assembly **100** as the dissector tip **120** is moved between a first configuration (FIG. 8) in which the dissector tip **120** extends in a direction substantially parallel to the longitudinal axis “X” of the tool assembly **100** and a second configuration (FIG. 9) in which the dissector tip **120** of the tool assembly **100** extends in a direction to define an acute angle Ω with the longitudinal axis “X” of the tool assembly **100**. In certain aspects of the disclosure, a distal portion **112a** of the cartridge assembly **112** defines a tissue guide surface **160** that defines an angle θ with the longitudinal axis “X” of the tool assembly **100**. In order to move the dissector tip **120** between the first and second configurations, the tip beam **140** and the tip beam cover **142** are manually grasped by a clinician and bent or deformed in the direction of arrows “A” or “B” in FIGS. 8 and 9, respectively, to a desired configuration best suited to perform a certain surgical procedure e.g., in which angle Ω is from about zero to about 45 degrees. The tip beam **140** is formed of a material that will retain its deformed configuration as the dissector tip **120** is used to separate target tissue from body tissue as the tool assembly **100** is advanced endoscopically to a surgical site.

(25) FIGS. 10-14 illustrate other exemplary aspects of a tool assembly **200** according to the disclosure. The tool assembly **200** includes an anvil assembly shown generally as anvil assembly **214**. The anvil assembly **214** is similar to the anvil assembly **114** but includes modifications to the dissector tip **120**. Only those modifications to the dissector tip shown generally as dissector tip **220** will be described in detail herein.

(26) The anvil assembly **214** includes a base portion **216**, an anvil portion **218**, and the dissector tip **220**. The base portion **216** includes a proximal mounting portion **216a** and a longitudinal rib **216b** that extends distally from the proximal mounting portion **216a** towards a distal end of the anvil portion **218**. The proximal mounting portion **216a** includes spaced through bores **216c** that receive pivot members **117** (FIG. 1) that pivotably couple the cartridge assembly **112** (FIG. 1) to the anvil assembly **214**. The anvil portion **218** is secured to the base portion **216** by, e.g., welding, and includes an upper surface **222** (FIG. 15), a proximal portion **218a**, and a distal portion **218b**. The upper surface **222** of the anvil portion **118** defines a plurality of rows of staple deforming pockets (not shown) that are positioned on opposite sides of a knife slot **228** (FIG. 13). The distal portion **218b** of the anvil portion **118** includes a clevis **230** that defines spaced openings **234** that receive a clevis pin **236**. The distal portion **218b** of the anvil portion **218** includes a distal face **238** that has a protrusion **240**.

(27) The dissector tip **220** has a body including a proximal portion **220a** and a distal portion **220b**. The proximal portion **220a** includes a hinge portion **250** that defines a through bore **252** that

receives the clevis pin **236** to pivotably secure the dissector tip **220** to the distal end portion **218b** of the anvil portion **218**. The dissector tip **220** has a proximal face **254** that defines spaced recesses **256a** and **256b** that are positioned to receive the protrusion **240** on the distal face **238** of the anvil portion **218** (FIG. 15). The dissector tip **220** has a tapered distal face **260** in the distal direction (FIG. 15) such that the distal end of the dissector tip **220** has a height that decreases in the distal direction. In addition, the width of the dissector tip **220** may also decrease in the distal direction (FIG. 11.) The dissector tip **220** has a through bore **261** that extends through the tapered distal face **260**. The through bore **261** allows a clinician to lasso or grasp the dissector tip **220** with a suture to allow the clinician to more accurately move and position the tool assembly **200** within a body cavity.

(28) FIGS. 15 and 16 illustrate the dissector tip **220** in a first configuration (FIG. 15) and in a second configuration (FIG. 16). In the first configuration (FIG. 15), the dissector tip **220** defines a longitudinal axis that defines an acute angle Ω with the longitudinal axis “X” of the tool assembly **200**. As discussed above, it is envisioned that angle Ω can be substantially the same as angle θ defined by the tissue guide surface **160** of the cartridge assembly **112**. In the second configuration, the dissector tip **220** defines a longitudinal axis that is substantially parallel to the longitudinal axis “X” of the tool assembly **200**. In order to move the dissector tip **220** between the first and second configurations, a clinician can apply a force to the dissector tip **220** to disengage the protrusion **240** from one of the respective spaced recesses **256a** and **256b** and pivot the dissector tip **220** about the clevis pin **236** to position the protrusion **240** in the other recess **256a** or **256b** of the dissector tip **220**. For example, when the dissector tip **220** is moved from the first configuration (FIG. 15) to the second configuration (FIG. 16), the clinician applies a force on the dissector tip **220** in the direction indicated by arrows “K” to pivot the dissector tip **220** about the clevis pin **236** in the direction indicated by arrow “L” to remove the protrusion **240** from the recess **256b** and move the protrusion into the recess **256a**. Although only two configurations are shown in the figures, it is envisioned that the dissector tip **220** may be moved to multiple positions. In that respect, it is envisioned that two or more spaced recesses **256** can be provided to receive the protrusion **240** to retain the dissector tip **220** in multiple different angular positions in relation to the longitudinal axis of the tool assembly **200**.

(29) Although the drawings only show retaining structure including a protrusion and a plurality of recesses, it is also envisioned that a variety of different types of retention structures can be provided to releasably retain the dissector tip in different angular positions. Further, the retaining structure can be supported on or integrally formed with either or both of the dissection tip **220** and the anvil assembly **214**.

(30) Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary aspects of the disclosure. It is envisioned that the elements and features illustrated or described in connection with one exemplary embodiment may be combined with the elements and features of another without departing from the scope of the disclosure. As well, one skilled in the art will appreciate further features and advantages of the disclosure based on the above-described aspects of the disclosure. Accordingly, the disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims.

Claims

1. A surgical stapler comprising: an elongate body having a proximal portion and a distal portion; and a tool assembly defining a longitudinal axis and including: a cartridge assembly; an anvil assembly including a proximal portion and a distal portion, the distal portion of the anvil assembly having a support bracket, the anvil assembly coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position; and a dissector tip including a

body having a proximal end and a distal end, the body having a thickness that decreases towards the distal end, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, wherein the dissector tip includes a tip beam and a tip beam cover, the tip beam including a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the deformable distal beam portion being formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the deformable distal beam portion in the desired configuration absent an external force applied to the dissector tip.

2. The surgical stapler of claim 1, wherein the proximal portion of the tip beam is secured to the support bracket of the anvil assembly by welding.

3. The surgical stapler of claim 1, wherein the tip beam cover is formed of a biocompatible material selected from the group consisting of polymeric materials, metals, ceramics, and elastomeric materials.

4. The surgical stapler of claim 1, wherein the dissector tip has a width that decreases towards the distal end of the dissector tip.

5. A tool assembly comprising: a cartridge assembly; an anvil assembly including a proximal portion and a distal portion, the distal portion of the anvil assembly having a support bracket, the anvil assembly coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position, the tool assembly defining a longitudinal axis; and a dissector tip including a body having a proximal end and a distal end, the body having a thickness that decreases towards the distal end, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, wherein the dissector tip includes a tip beam and a tip beam cover, the tip beam including a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the deformable distal beam portion being formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the tip beam portion in the desired configuration absent an external force applied to the dissector tip.

6. A tool assembly comprising: a cartridge assembly; an anvil assembly including a proximal portion and a distal portion, the anvil assembly including a base portion and an anvil portion, the base portion coupled to the cartridge assembly such that the tool assembly is moveable from an open position to a clamped position, the anvil portion secured to the base portion and including a plurality of staple deforming pockets, the anvil portion including a distal portion including a support bracket, the tool assembly defining a longitudinal axis; and a dissector tip including a body having a tapered distal surface and a thickness that decreases in the distal direction, the body being movably coupled to the anvil assembly for movement between a first configuration substantially aligned with the longitudinal axis and a second configuration defining an acute angle with the longitudinal axis, the dissector tip including a tip beam and a tip beam cover, the tip beam having a proximal portion and a deformable distal beam portion, the proximal portion of the tip beam fixedly secured to the support bracket, the tip beam cover received about the tip beam and the support bracket and having an atraumatic configuration, the proximal portion of the dissector tip secured to the bracket and the deformable distal beam portion formed of a material that is manually bendable by a clinician to any desired configuration including and between the first and second configurations, the material having properties to retain the distal beam portion in the desired configuration absent an external force applied to the dissector tip.

